Progressing Towards a Label-Free Live-Cell Kinase Assay

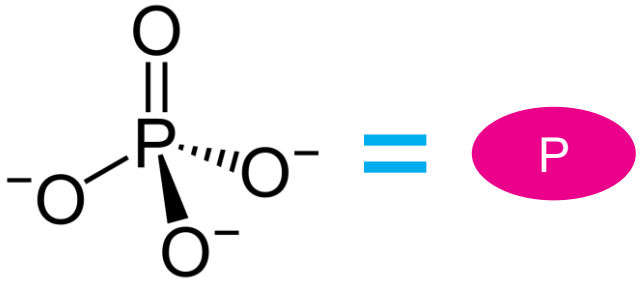
Joshua Schaaf

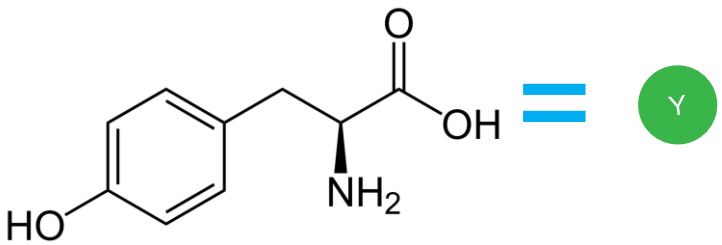
LFT Days, Turku, 13.02.2026

Introduction & Background

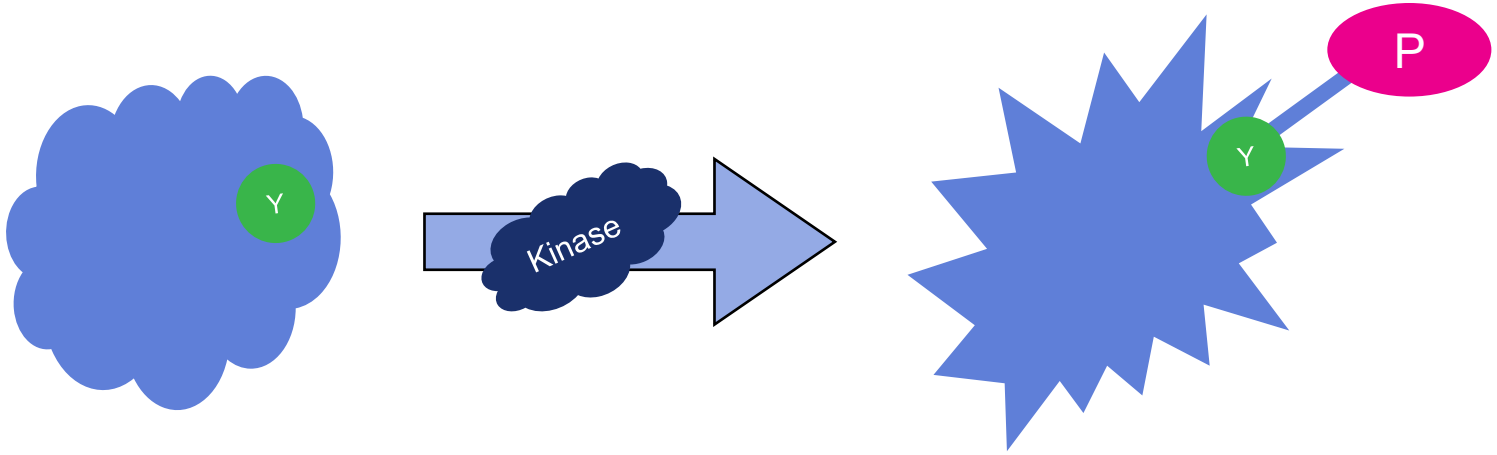
how are we planning to measure kinase activity?

Kinases and Phosphorylation



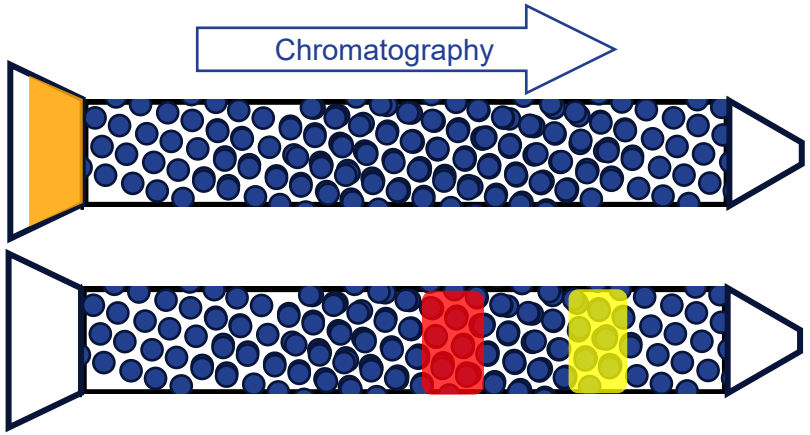


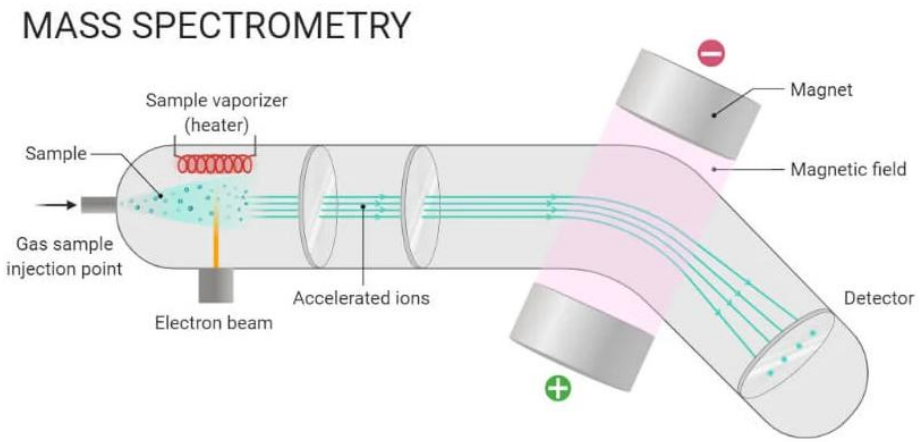
- Proteins perform most of the ‘work’ in life
- Kinases add phosphates to proteins
- Phosphate presence changes protein ability to perform cellular tasks
  - Changes in kinase activity are **linked to many diseases e.g. cancer**



How are molecules measured?

- We must know some molecular attributes
  - Weight, size, charge, resonance, reactivity, vibrational modes: (consequences of **composition and structure**)
- We measure the attribute(s) which provide us with an accepted level of accuracy
- **Phosphorylation** of proteins with LC/MS

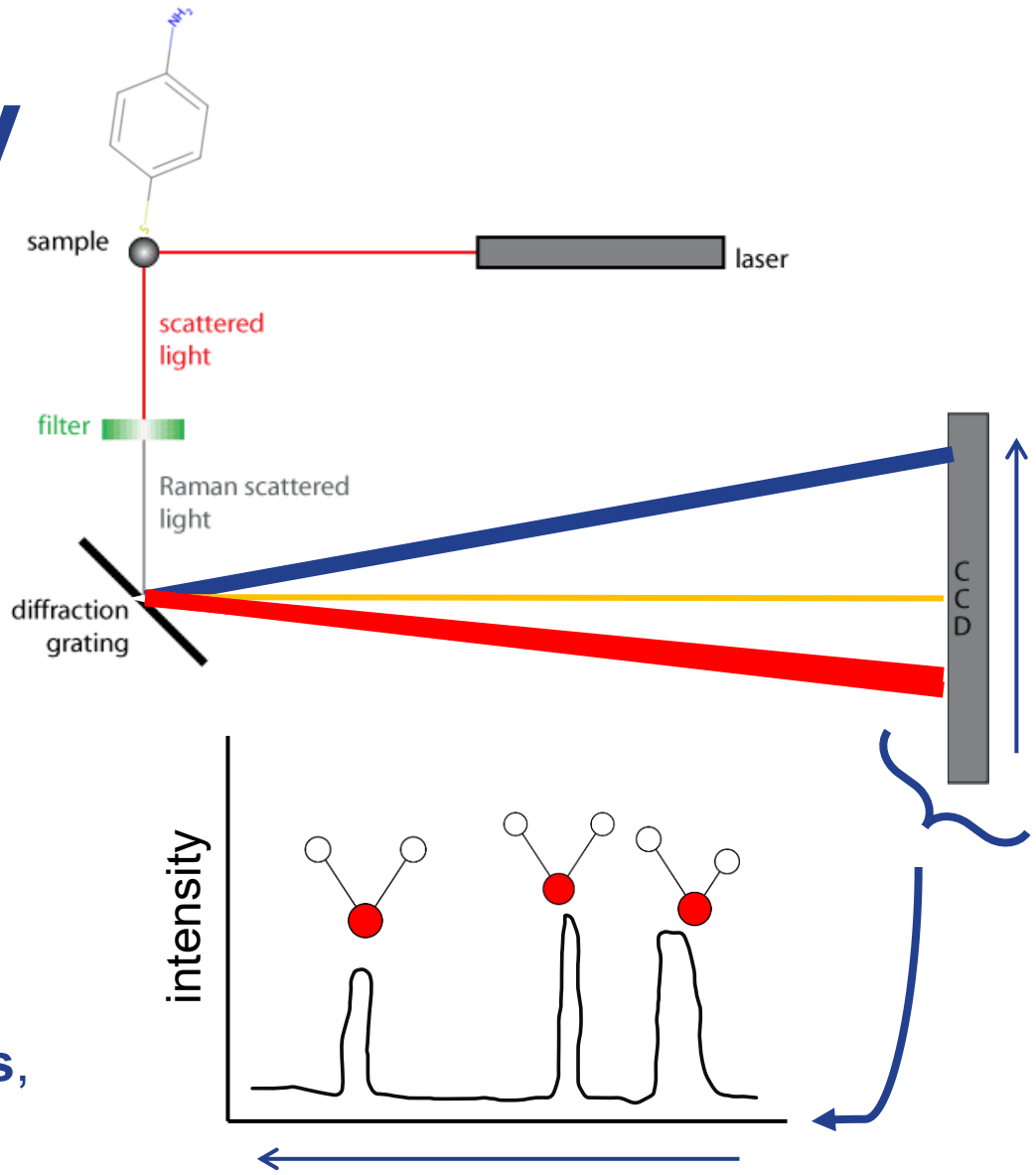




https://specac.com/theory-articles/molecular-vibrations-explained-animated-guides/?cst
https://microbenotes.com/mass-spectrometry-ms-principle-working-instrumentation-steps-applications/

Raman Spectroscopy

- Illumination of samples with laser light, and measuring the light scattered back
- Scattered light is in different color(s) (wavelength shift)
- Wavelength shifts are attributed to different molecular vibrations
  - Specificity from unique molecular **vibrational modes**, i.e. a combination of separate wavelength shifts (“**molecular fingerprint**”)



https://www.doitpoms.ac.uk/tlplib/raman/method.php

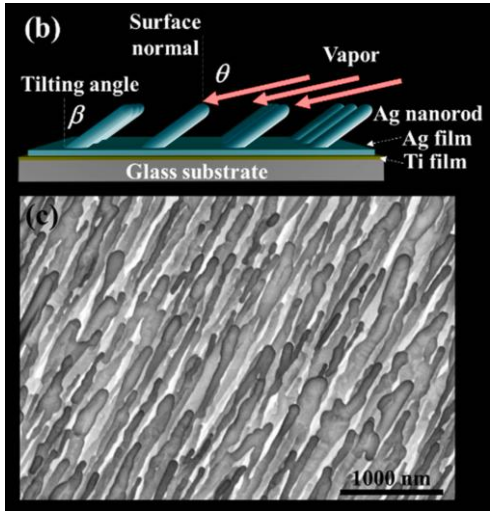
Raman Spectroscopy – Considerations

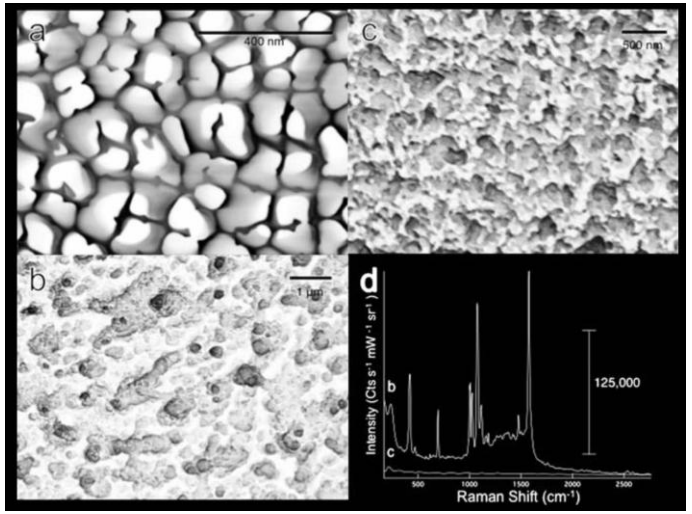
- Non-destructive (low enough laser power) and label-free
- Molecular fingerprinting provides **high specificity**
- Raman spectra are not ‘solved’ *
  - **Database(s) of spectra** must be created in the lab
- Less than 1% of the laser light is inelastically scattered
  - Needing **high power density** & **long acquisition times**

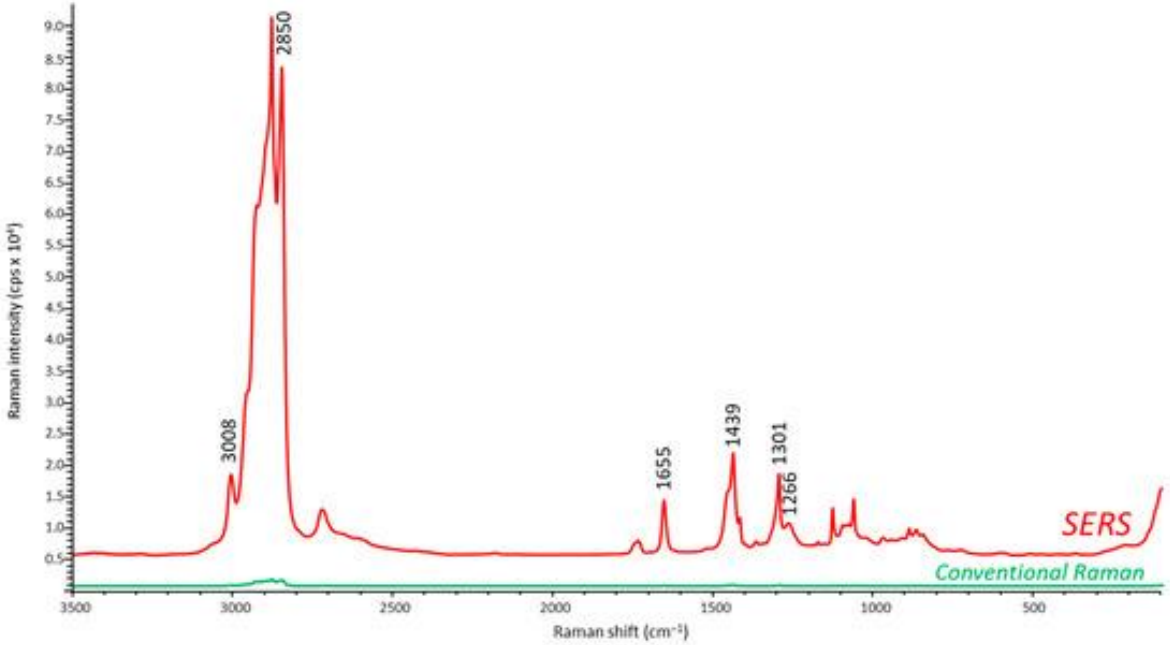
* Molecular dynamic (MD) simulations predicting Raman spectra exist, but only for the most simple cases

Surface Enhanced Raman Spectroscopy (SERS)

- Solves low inelastic scattering problem with **Raman-effect enhancement**
  - Nobel metal (Au, Ag, Pt) **nanostructures** provide enhancement factors (EF) through a combination of their electromagnetic & chemical properties (probably)
  - **Single molecule sensitivity** with lower laser power and shorter integration time



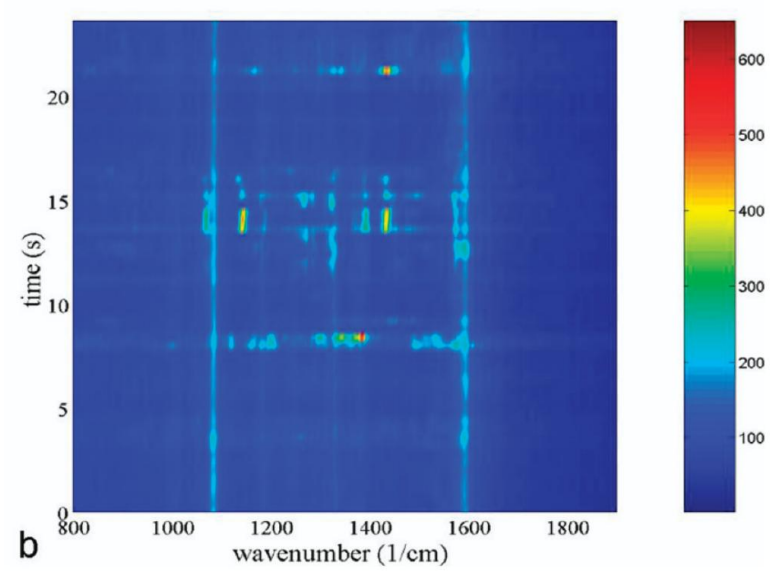


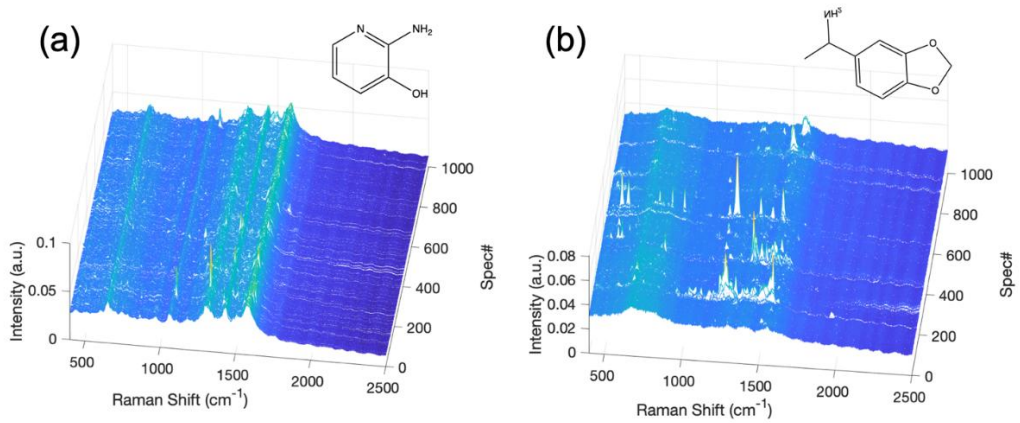


https://pubs.rsc.org/en/content/articlelanding/2024/cs/d3cs00540b
https://pubs.rsc.org/en/content/articlelanding/2011/an/c1an15432j

SERS – Considerations

- Requires more research / databasing…
  - SERS substrate composition, nanostructure, different interactions of substrate and sample – **all influence spectra**
- Biological samples are complex, even with high sensitivity
- Enhancement regions are only 10s of nanometers in size
  - Biomolecules can be huge in comparison

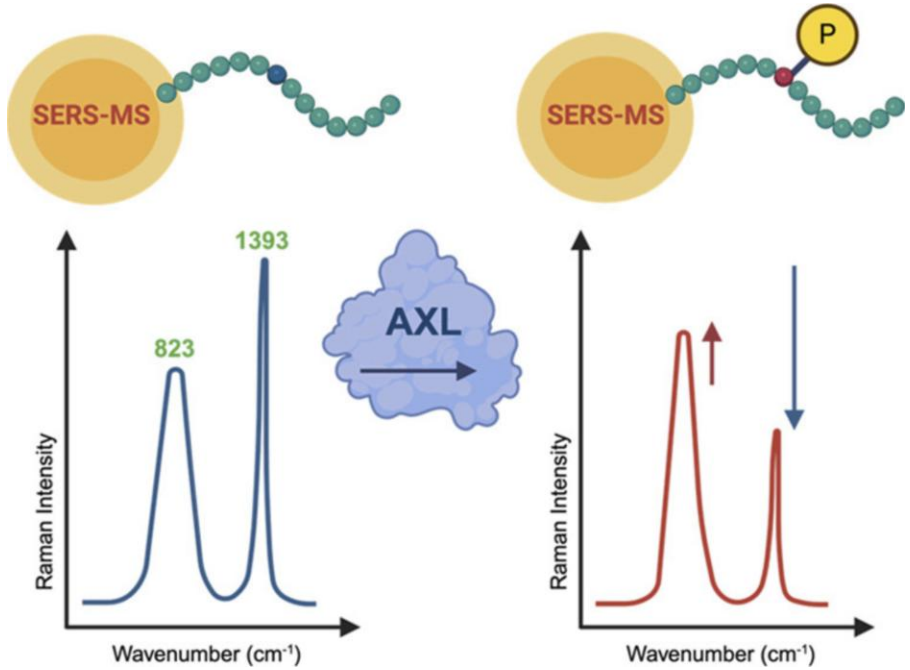


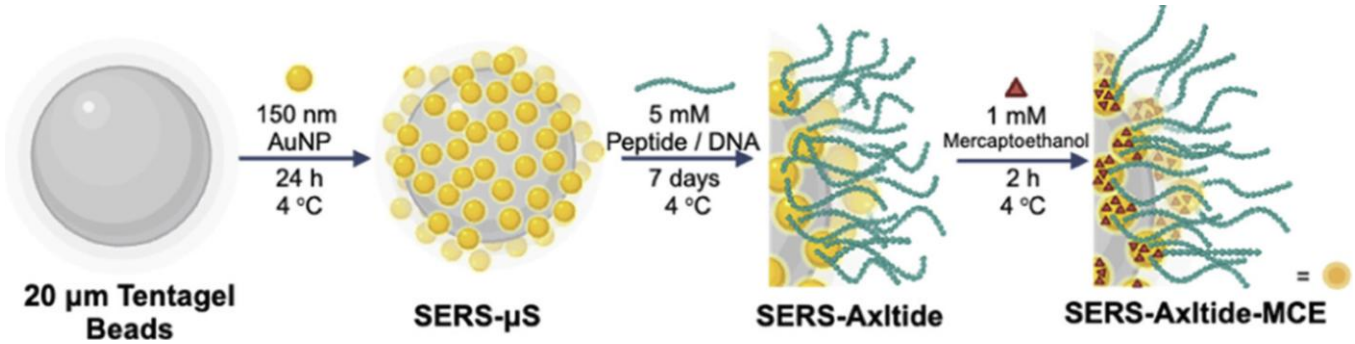


https://doi.org/10.1021/jp100861w
https://pmc.ncbi.nlm.nih.gov/articles/PMC7028501/

Bioconjugated SERS Substrates

- Reduces complexity, increases specificity
- Biomolecules attached to SERS substrates
  - **Gold–sulfur bond (Au–S)** well established in literature
  - **Cysteine**, alkanethiols, 2-mercaptoethanol, etc
- Target molecule measured via **interaction with conjugated biomolecules**
  - Phosphorylation of a peptide by a kinase





https://doi.org/10.1021/acs.analchem.4c02860

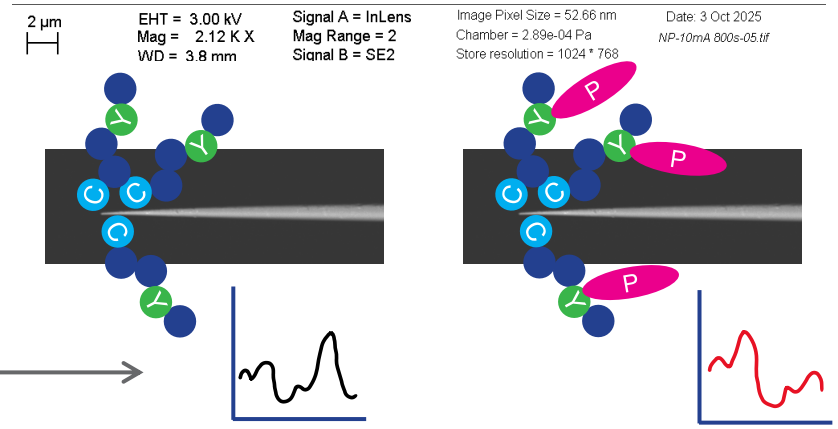
Current Progress

testing SERS substrates

Plan for Measuring Phosphorylation

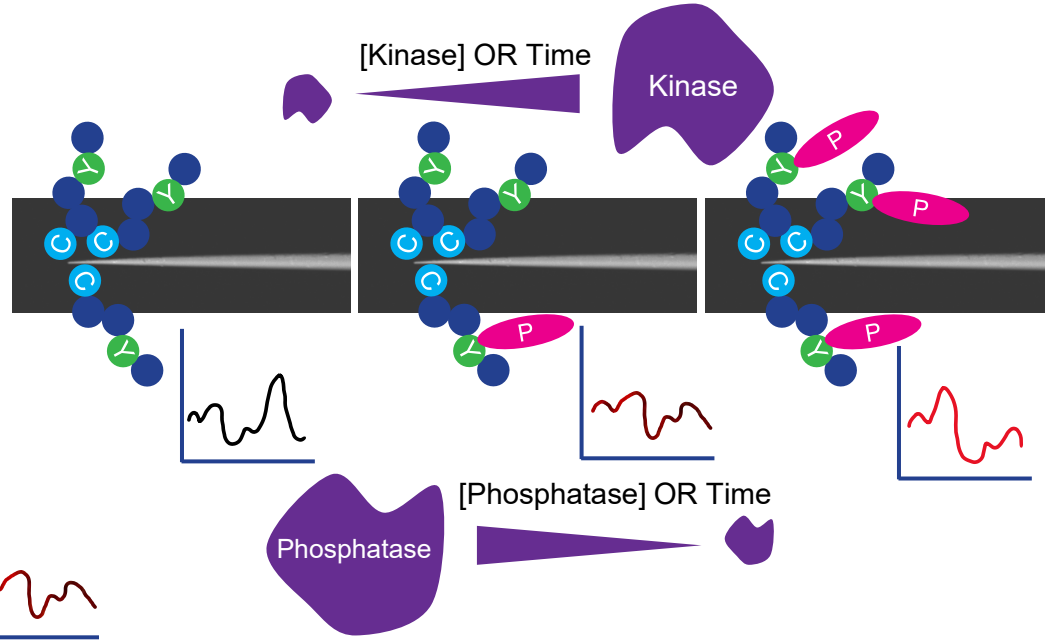
1. **Attach peptides** to nanoneedle
2. **Measure differences** between phosphorylated and non-phosphorylated peptides
   - Dried, in solution (buffer), in cell
3. **Measure changes in real-time** kinase-phosphorylated, phosphatase-dephosphorylated
   - In solution (buffer), in cell (goal!)

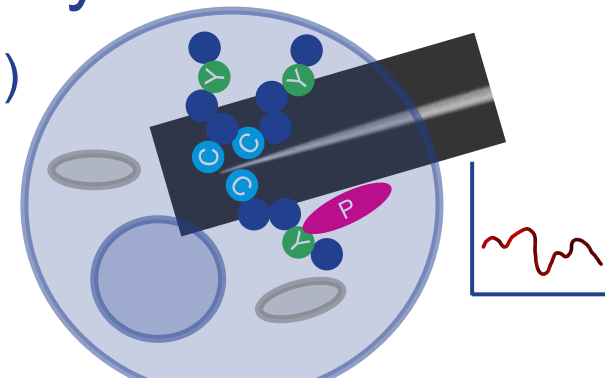
1.



2.

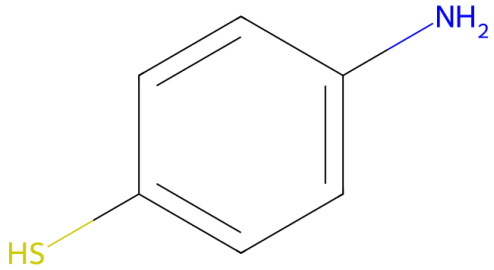
3.

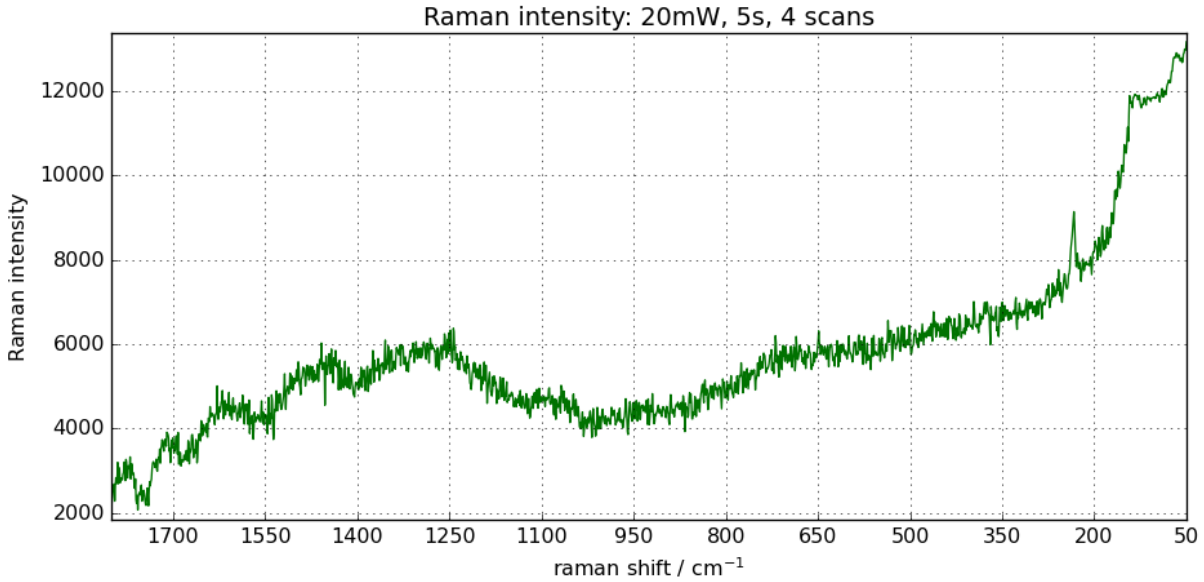




Testing SERS Nanoneedle Signal

- Attached 4-aminothiophenol (4-ATP) to SERS substrate and measured Raman signal

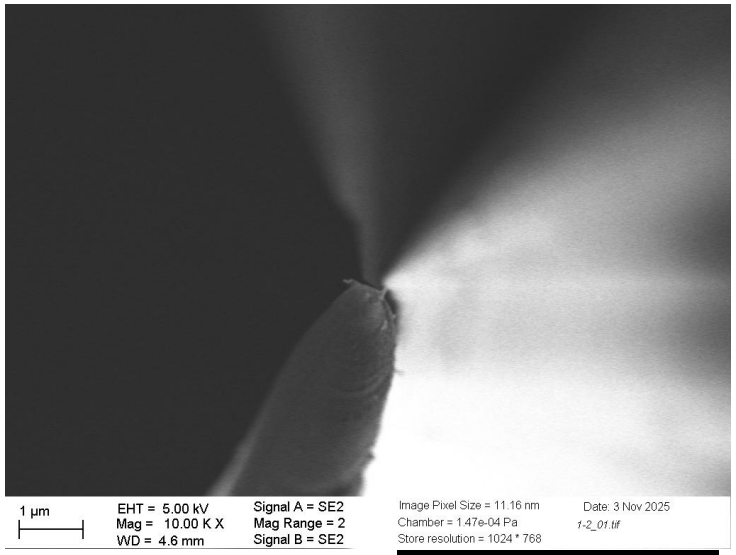


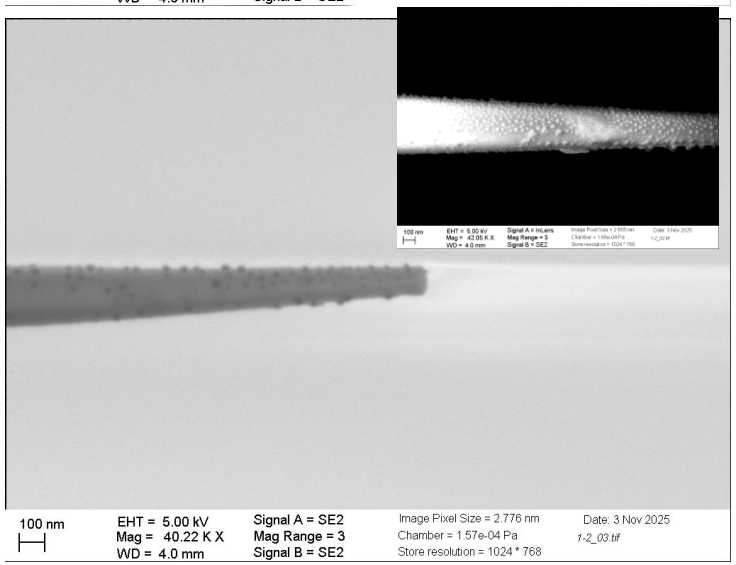


- **No 4-ATP peaks identifiable**

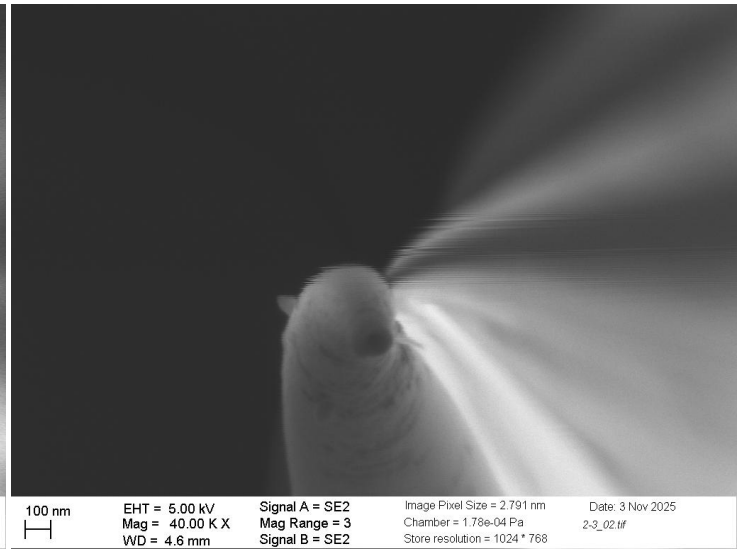
Checking Structure – Nanoneedle

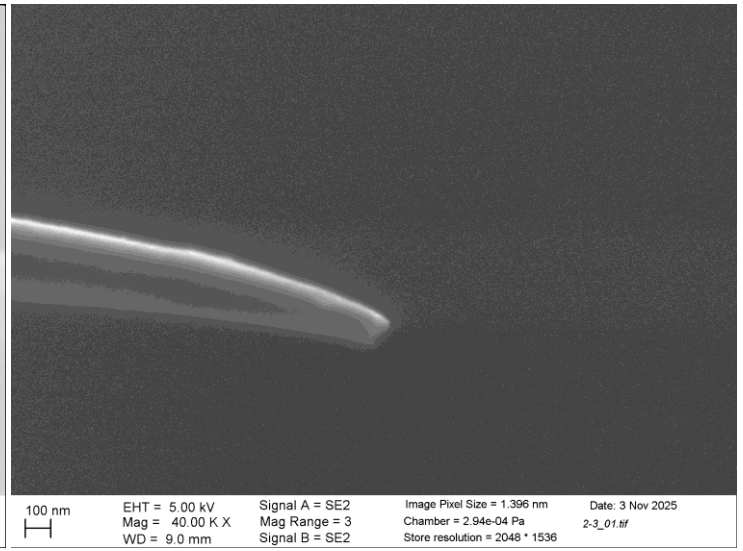
1





2





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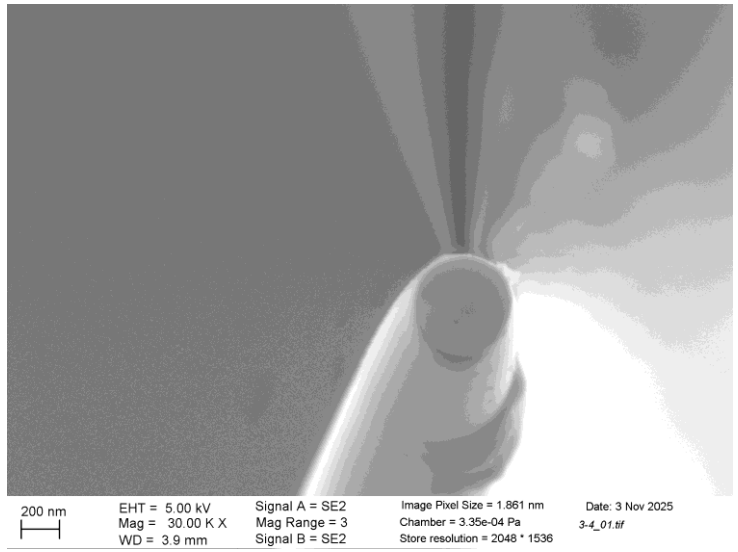
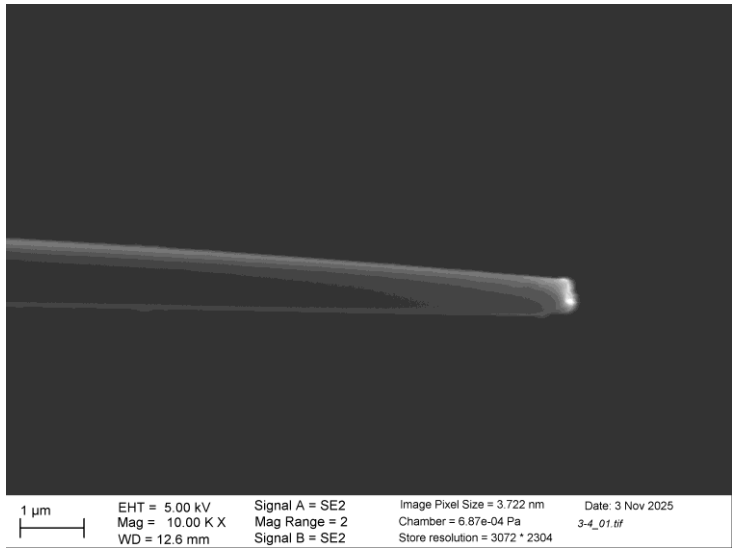


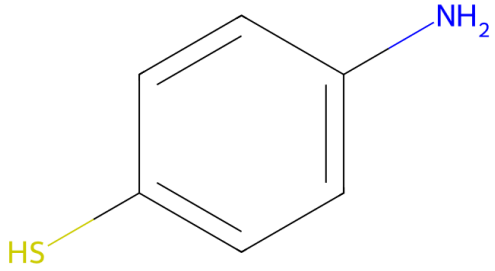


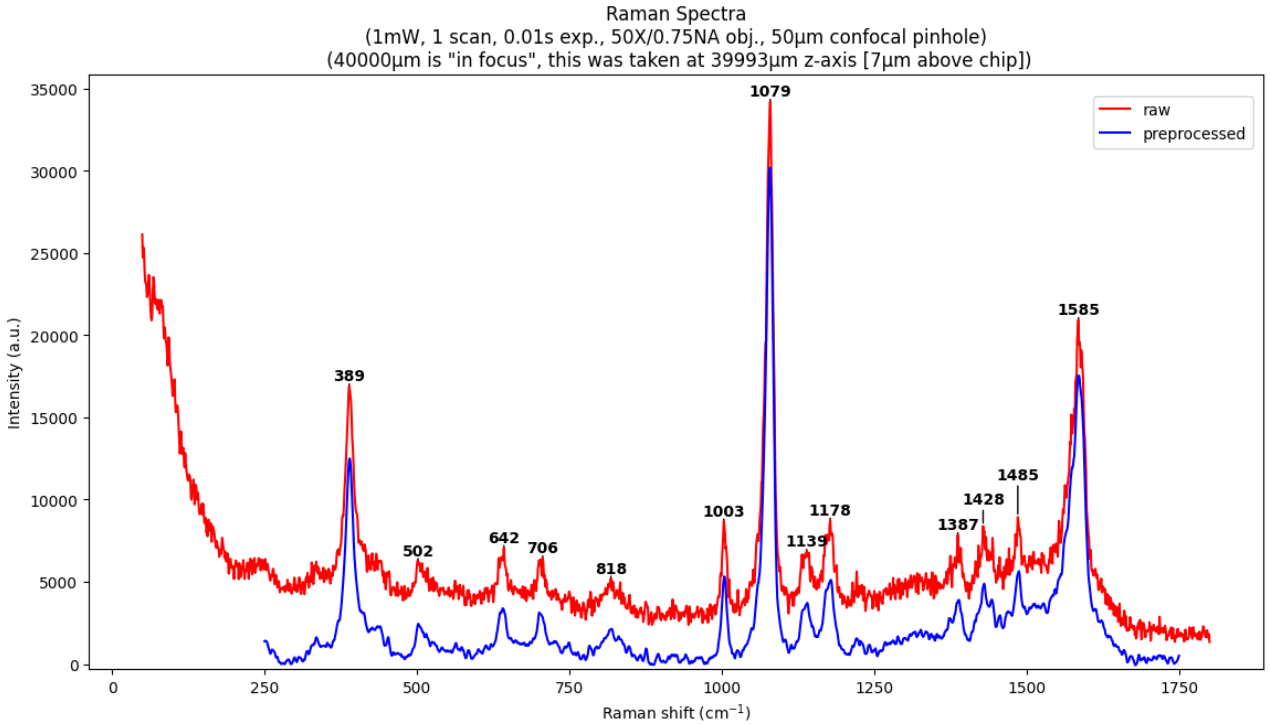
⌀ ???????	Inner diameter	⌀ 23px (64.19nm)	Inner diameter	⌀ 152px (282.87nm)	
⌀ 39px (108.26nm)	Outer diameter	⌀ 82px (114.47nm)	Outer diameter	⌀ 244px (535.96nm)	

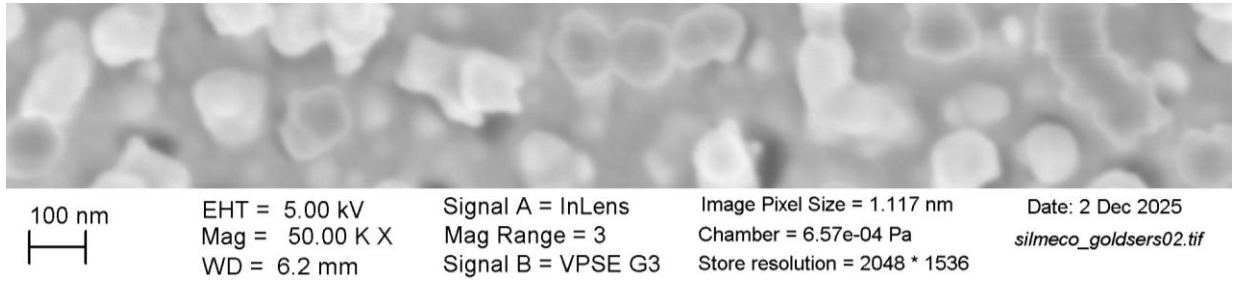
Commecial SERS Substrate Signal

- Continue research in the meantime by using Silmeco's Gold SERStrate

- **Strong 4-ATP signal**

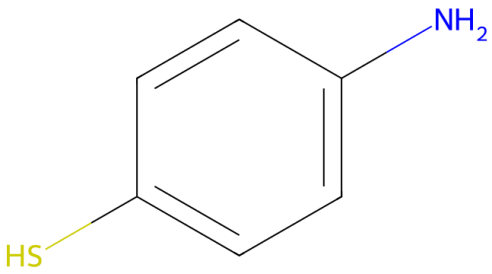


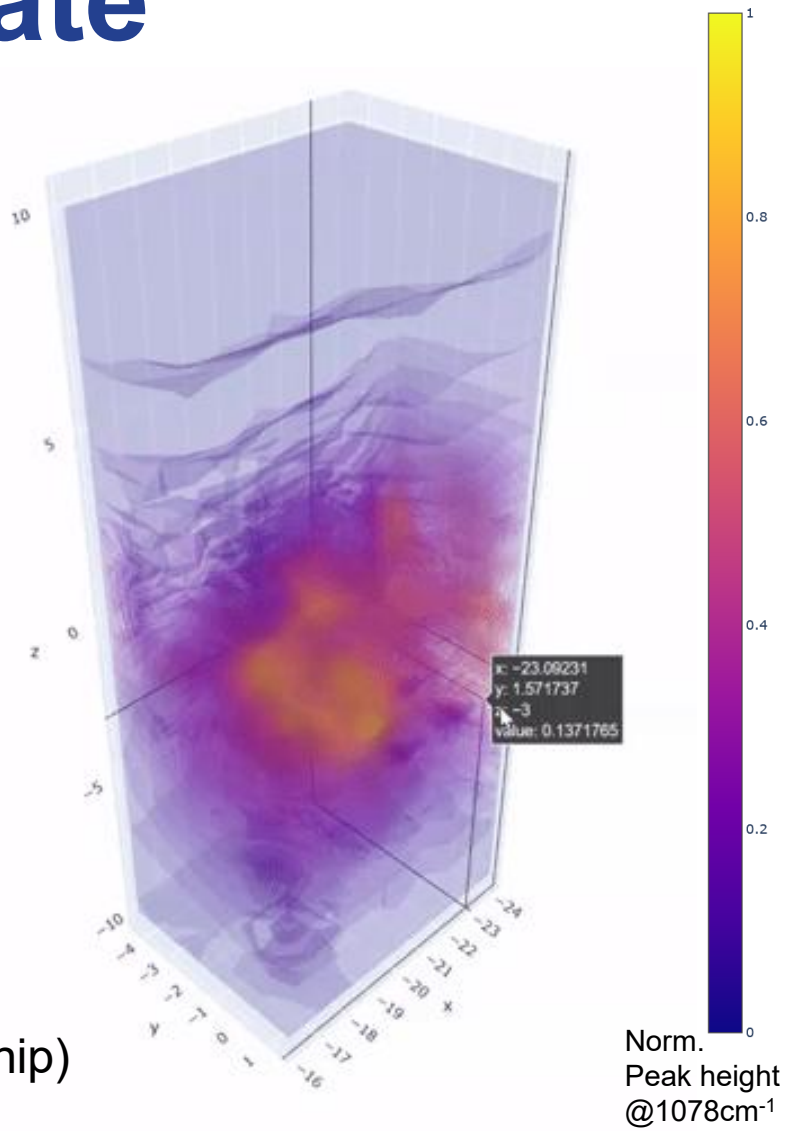




Commecial SERS Substrate

- Measured 8 x 5 x 20 µm volume
  - Confocal microscopy
- Signal **varied spatially**
  - SERS enhancement is varied spatially
  - 4-ATP molecules are not bound evenly across substrate





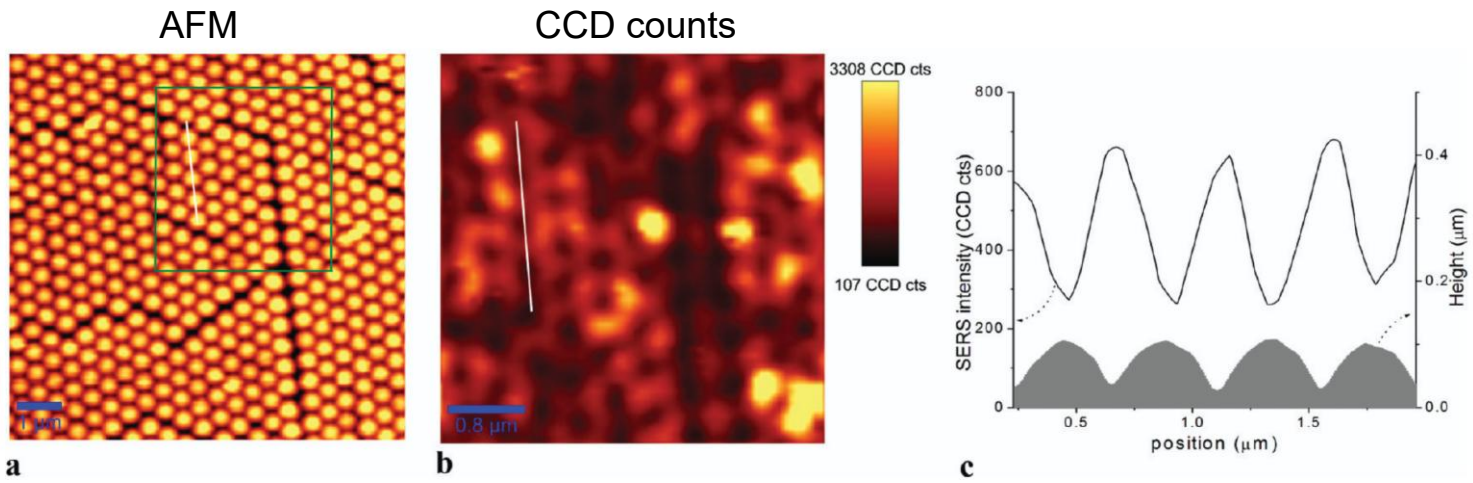
z = 0 is ”in focus”
(negative is into the chip)

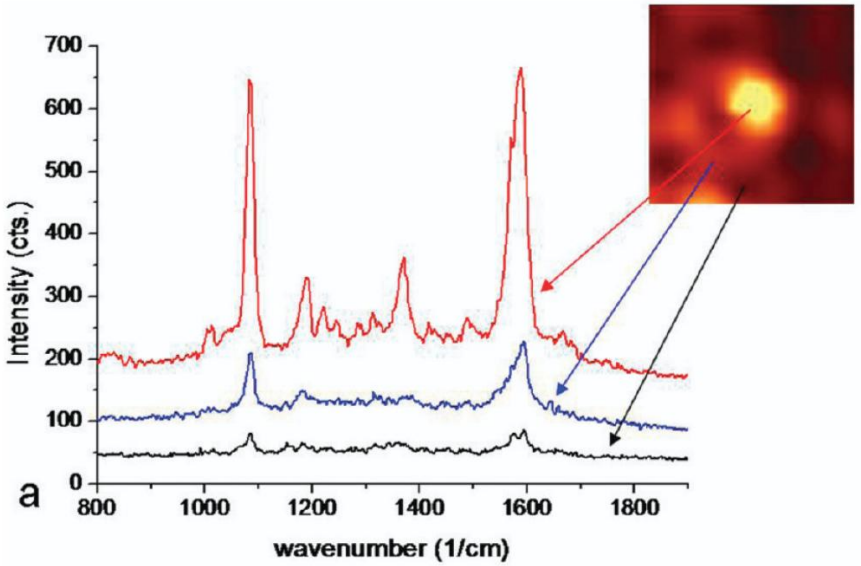
Norm.
Peak height
@1078cm^{-1}

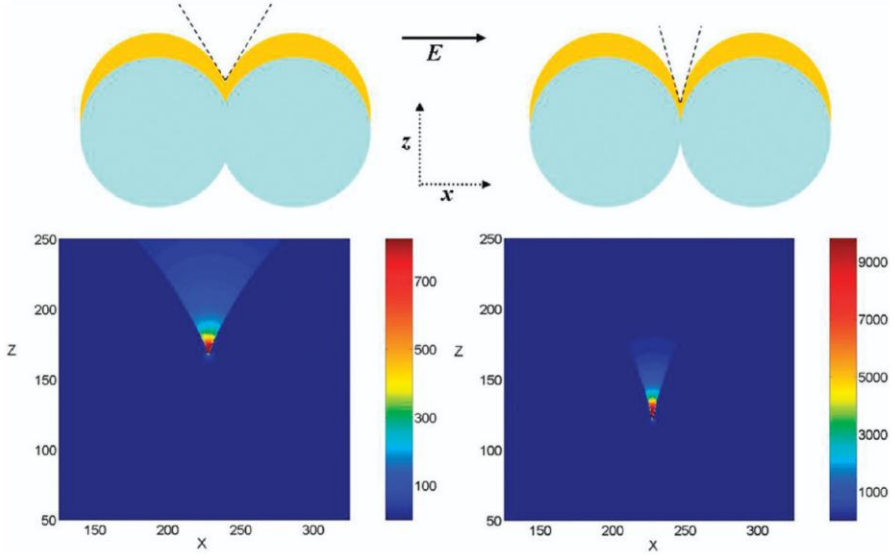
x, y, and z in µm

Spatial Variation – Literature Check

- SERS substrates have repeatedly shown spatial variation of enhancement
  - This is expected!





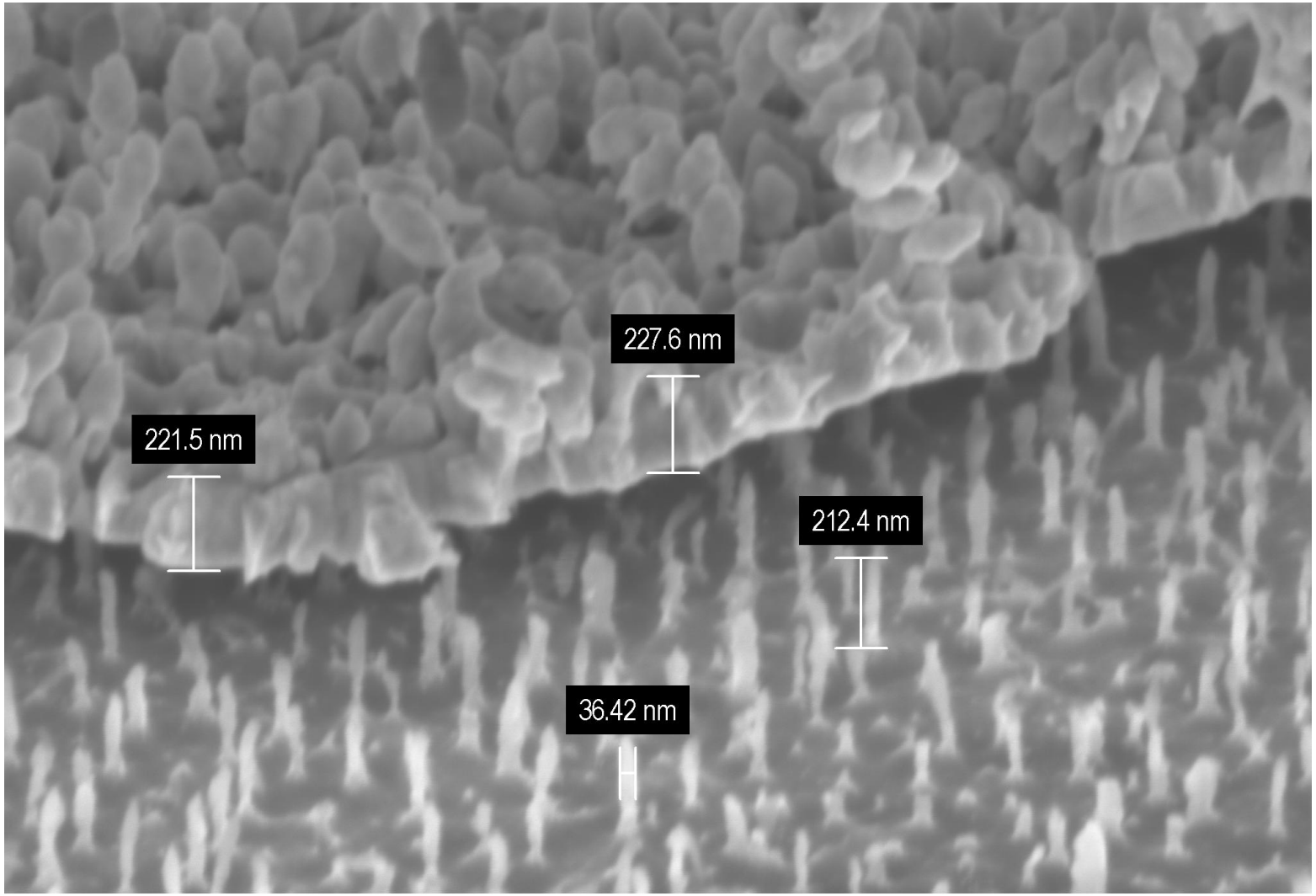


https://doi.org/10.1021/jp100861w

Checking Structure – SERStrate

Corner of Silmeco Gold SERS substrate SEM

View from angle (52 degrees)



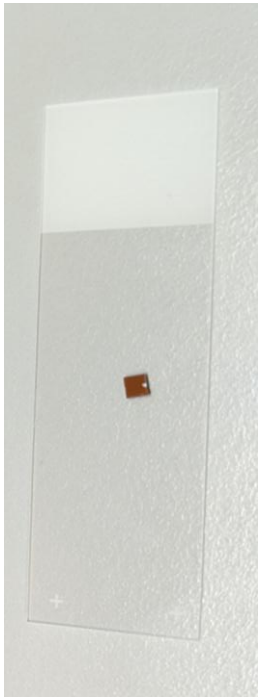
100 nm

EHT = 5.00 kV
Mag = 36.79 K X
WD = 4.1 mm

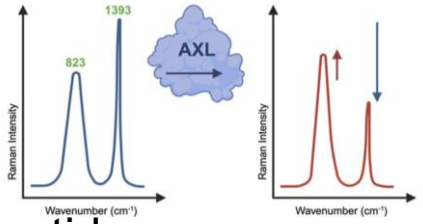
Signal A = InLens
Mag Range = 3
Signal B = VPSE G3

Image Pixel Size = 1.517 nm
Chamber = 2.31e-04 Pa
Store resolution = 2048 * 1536

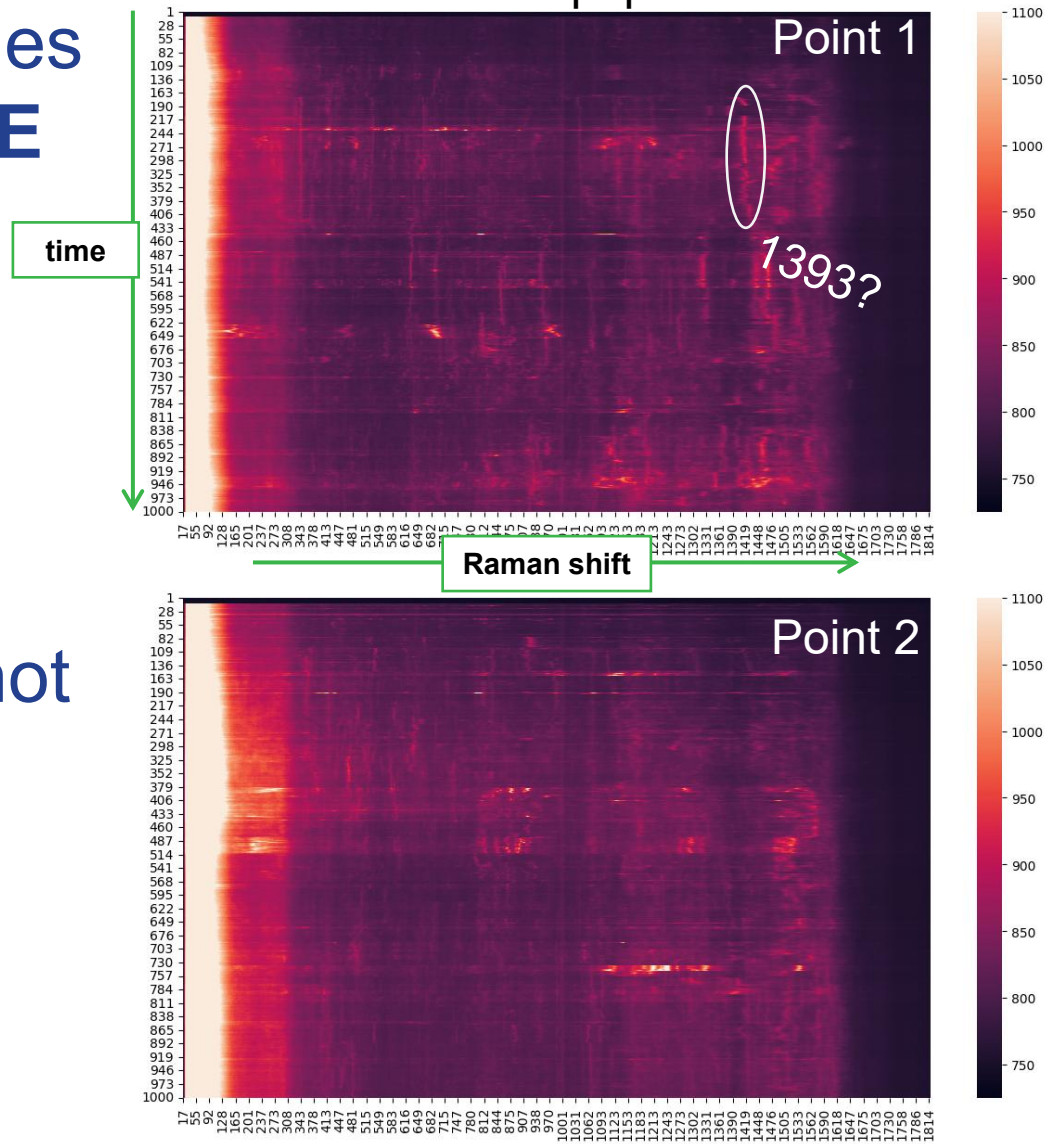
Date: 2 Dec 2025
silmeco_goldsers-tilted-06.tif

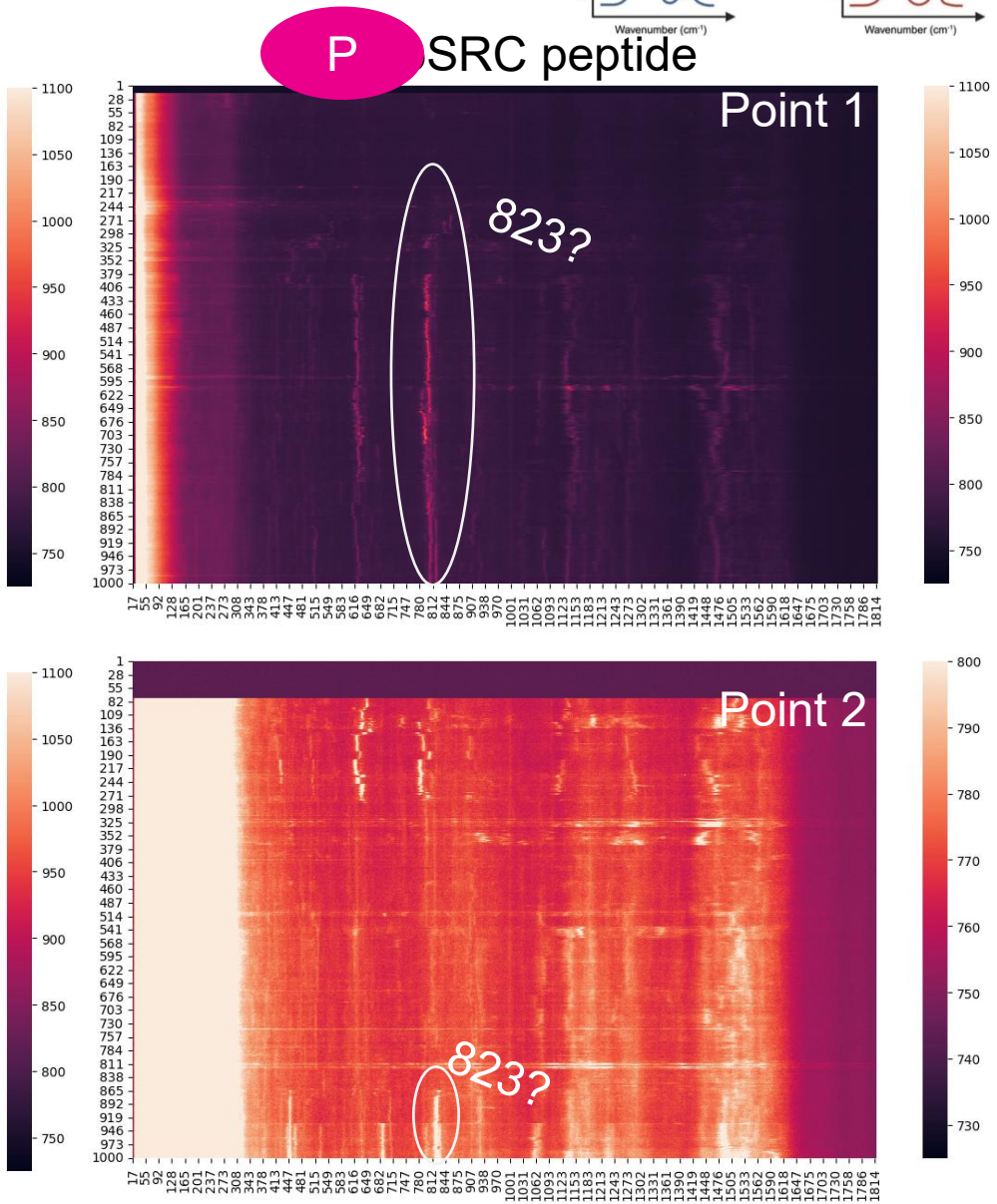


Measure Peptide Signal



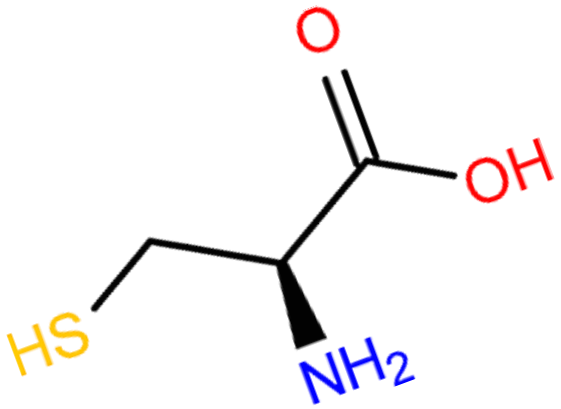
- 16-mer peptides (**AEEEIYGEFE AKKKKC**)
- Signal varies **spatially** and **temporally**
- Many peaks not immediately assignable to peptides

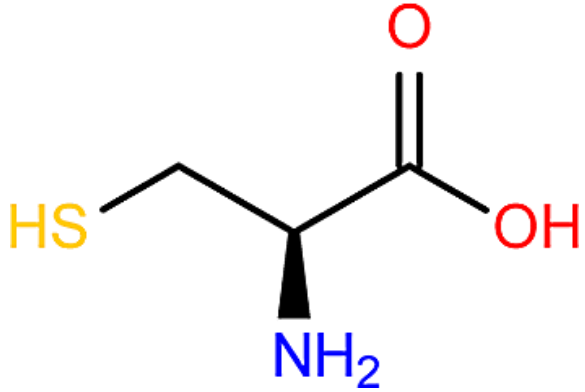


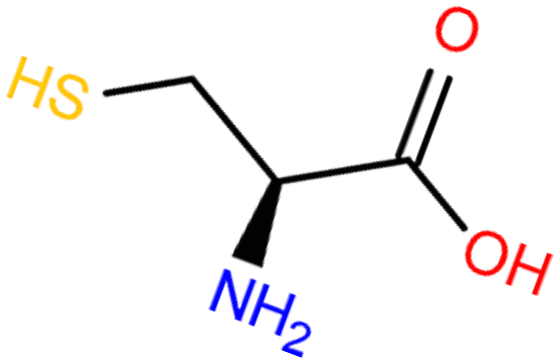


Next Steps

- Re-measure with **lower laser power**
  - Avoid substrate melting / molecule burning with lower power density
- Find the **limit of detection** with **cysteine** alone
  - Binds to substrate in a predictable way (fewer expected peaks compared to peptide)

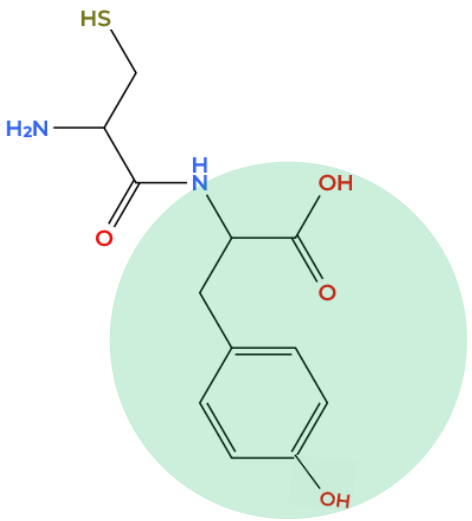


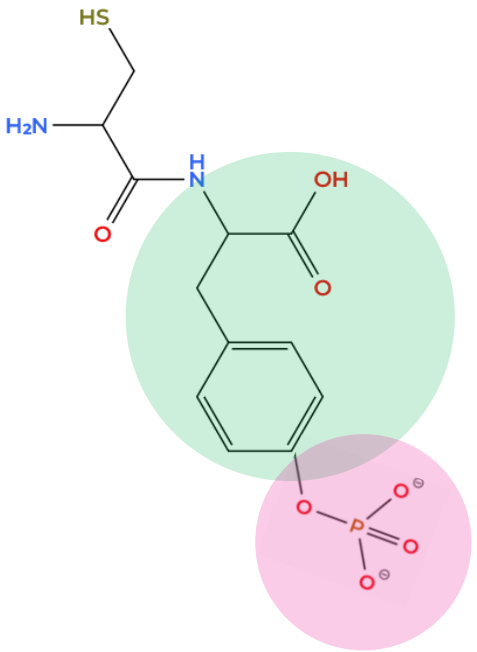


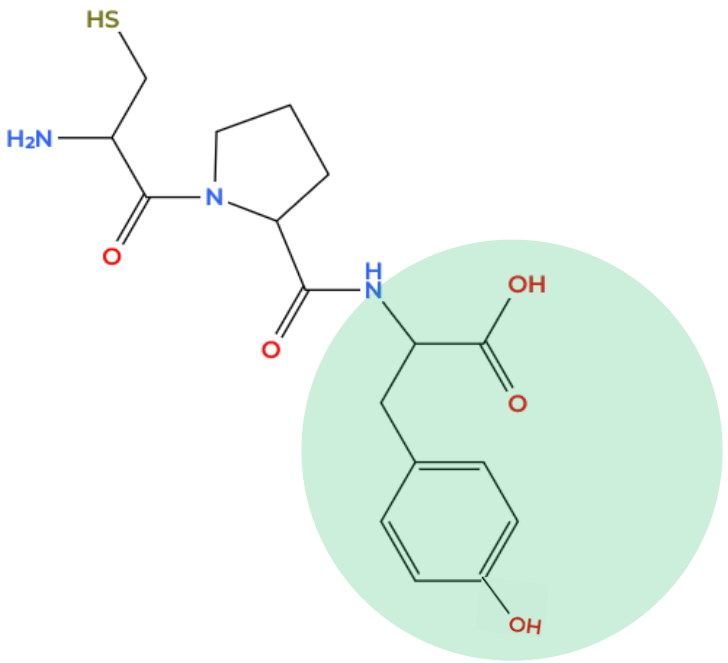


Next Steps?

- Measure cysteine-tyrosine dimers, w/ & w/o phosphate
- Use peptides of different lengths to determine our ability to measure signal further from the substrate







JANE AND AATOS ERKKO FOUNDATION

Kiitos

PROGRESSING TOWARDS A LABEL-FREE LIVE-CELL KINASE ASSAY

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Kysymyksiä?